Department of Computer Science and Engineering

University of Barishal

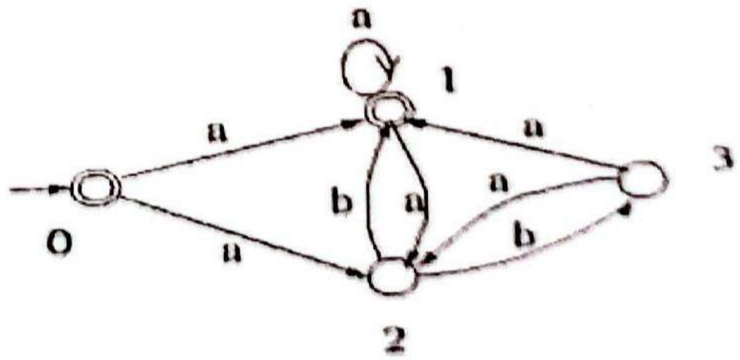
Mid-Term Exam (16 June 2026)

Course: Theory of Computation | Marks: 60 | Time: 60 Min

Instructor: Dr. Md Manjur Ahmed , Associate Professor

Questions:

1. Prove that every language defined by a regular expression is also defined by a finite automata. [10 Marks]
2. Prove that if L is a regular language over alphabet Σ, then $\bar{L} = \Sigma* - L$ is also a regular language. [10 Marks]
3. Convert the following NFA to DFA. [10 Marks]



4. Convert the following RE to ε-NFA: (0+1)*01(0+1)* [10 Marks]
5. Why explicit ε-transitions in finite automata is important? [10 Marks]
6. Build an ε-NFA for the following language: [10 Marks]
   L = {w | w is empty, or if non-empty will end in 01)

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